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Studierichting: Informatica
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$$\int \frac{x^2}{\sqrt{1-x^2}}dx$$

Stel $x = \sin(t) \Rightarrow dx = \cos(t)dt$

$$= \int \frac{\sin^2(t)}{\cos(t)} \cos(t)dt$$

$$= \int \sin^2(t)dt$$

$$= \int \frac{1-\cos(2t)}{2}dt = \frac{1}{2}t - \frac{1}{4}\sin(2t) + c$$

$$= \frac{1}{2}\operatorname{Bgsin}(x) - \frac{1}{2}\sin(t)\cos(t) + c$$

$$= \frac{1}{2}\operatorname{Bgsin}(x) - \frac{1}{2}x\sqrt{1-x^2} + c$$

References